Pre-work · Cell Anatomy

Cell Anatomy. Read the Part A chapter first, then work this in order. You come to class ready to use this, not to hear it for the first time.

Module 1 · 2 of 4 | Bring this to class

How to work this pre-work

The order matters more than the hours. Context first, content second. Most students who struggle did every piece of this, in the wrong sequence.

1. **Read** Read the Part A chapter first. This packet does not repeat it. It puts it to work.
2. **Organize** Panel 1. Turn the reading into mini visuals. Not copied sentences, your structure.
3. **Draw** Panel 2. Brain dump from memory, then draw, then correct in a second colour.
4. **Apply** Panel 3. Notes closed for the first pass. Open them only to check.
5. **Retrieve** Later in the week, not the same night. Cover everything and redraw your mini visuals from memory. What you cannot redraw is what you do not yet know.

Why this order. Reading before you organize gives you context. Organizing before you draw forces you to decide what connects to what. Drawing before you apply is where you find out what you actually know. If you skip to the application questions, you will answer them with the notes open and learn almost nothing.

PANEL 1 OF 3

Organize it

Turn the cell chapter into mini visuals. Eight chunks, eight small visuals, each in the shape that fits it.

FIVE SHAPES, THAT IS THE WHOLE TOOLKIT

Information has a shape. Pick the wrong one and the visual is noise. Pick the right one and you can redraw it from memory in twenty seconds. Ladder for an ordered series. Chain for a sequence, with arrows. Split for a pair you keep confusing. Hub for one thing with parts hanging off it. Grid for two variables crossed.

Your chunks for this packet

Each chunk gets one visual, in the shape that fits it, drawn on the mini visual sheet at the end of this packet. Each is made once, so nothing here is done twice. If a chunk

PANEL 2 OF 3

Draw it

Panel 1 was the shape of the information. This is the structure itself, drawn from memory, then corrected.

HOW A BRAIN DUMP WORKS

Set a timer for two minutes. Close everything, write and sketch every piece you can recall on blank paper. Do not organize it. When the timer stops, then draw. The gap between the dump and the drawing is the part that teaches you.

PANEL 3 OF 3

Apply it

First pass with everything closed. Then open the notes in a second colour and mark what you missed.

WORKED EXAMPLE

A drug destroys the enzymes inside lysosomes. Predict what accumulates and where.

Answer. Undigested macromolecules and worn organelles build up inside the cell.

Reasoning. Work from structure to consequence. The lysosome is the digestive organelle, so its job is breaking down macromolecules, damaged organelles, and pathogens. Remove the enzymes and nothing else takes over that job, so the material has nowhere to go and stays inside the cell. Cells with the highest turnover show it first. This is the mechanism behind Tay-Sachs, where one missing lysosomal enzyme lets lipid accumulate in nerve cells until the cell fails.

Set A · Structure to job

1. The organelle that assembles ribosomal subunits is the ________________________.

takes longer than three minutes you are copying instead of organizing.

1 HUB

The generalized cell

One hub, three spokes: plasma membrane, nucleus, cytoplasm. Hang one job off each spoke. This is the frame everything else attaches to.

2 GRID

Permeability

Semipermeable, permeable, impermeable down the side. Definition and one example across the top. Three rows, and the examples must be yours.

3 HUB

Inside the nucleus

Nucleus at the centre. Spoke to envelope, pores, chromatin, chromosomes, nucleoplasm, nucleolus. One job on each spoke.

4 SPLIT

Cytoplasm versus cytosol

One split. The trap is that one contains the other. Write the containing relationship on the branch, not just two definitions.

5 CHAIN

The secretory pathway

Nucleus, arrow, rough ER, arrow, vesicle, arrow, Golgi, arrow, vesicle, arrow, plasma membrane. Then write what happens to the protein at each arrow.

6 GRID

Rough ER against smooth ER

The two down the side. Structure, function, and what it sends onward across the top.

DRAWING 1

The generalized cell, labelled

BRAIN DUMP FIRST

Two minutes. Every organelle you can name and one job for each.

Draw one cell large enough to label. Include the plasma membrane, nucleus with nucleolus, rough and smooth ER, Golgi, mitochondria, lysosomes, peroxisomes, ribosomes, cytoskeleton, centrioles. Label every one. Do not draw a diagram you have seen, draw the one you remember.

In your notebook, head the page **M1–1 The generalized cell, labelled** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can point at any organelle on your drawing and name what fails if it stops working.

2. A cell that secretes large amounts of protein would be rich in ______________________.
3. A cell that detoxifies drugs would be rich in ______________________.
4. The organelle with its own DNA and ribosomes is the ______________________.
5. Name one cell type that is multinucleated and one that is anucleated.

____

Set B · Predict the failure

1. A mutation stops the cell producing glycoproteins. Predict two things the cell can no longer do well, and say why.

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2. A cell cannot make enough ATP. Name the organelle involved, and name the three tissues that would show it first. Explain your choice.

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3. The cilia lining the airway stop moving. Predict the clinical picture, and say which microtubule arrangement has failed.

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4. Cholesterol is removed from the membrane. Predict what happens to the membrane as temperature changes.

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Set C · Tell it apart

1. A student says cytoplasm and cytosol are the same thing. Correct them in one sentence.

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2. Free ribosomes and bound ribosomes make different proteins. Say what decides which is used.

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7 **HUB**

The cytoskeleton

Three spokes: microfilaments, intermediate filaments, microtubules. Protein on one line, job on the next. Circle the one that builds cilia.

8 **SPLIT**

9 plus 0 against 9 plus 2

Centrioles on one branch, cilia and flagella on the other. Write the arrangement and, underneath, the one word that separates them: movement.

YOU KNOW PANEL 1 WORKED WHEN

Someone hands you a blank page and you can redraw any one of these from memory in under a minute, explaining it out loud while you draw.

DRAWING 2

The plasma membrane, close up

BRAIN DUMP FIRST

One minute. Every component of the membrane and where it sits.

Draw a section of bilayer wide enough to show detail. Mark the hydrophilic heads and hydrophobic tails and which way each faces. Add an integral protein spanning the layer, a peripheral protein on the surface, cholesterol between the tails, and a glycoprotein with its sugar chain facing out. Label which side is extracellular.

In your notebook, head the page **M1–2 The plasma membrane, close up** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can explain why a lipid-soluble drug crosses this and a charged ion does not, using only your drawing.

3. Chromatin and chromosomes are the same material. Say what changes between them and when.

YOU KNOW PANEL 3 WORKED WHEN

Your first-pass answers and your corrections are in two different colours, and you can say out loud why each correction was a correction.

DRAWING 3

Trace the pathway on your cell

BRAIN DUMP FIRST

One minute. The route a secreted protein takes, in order.

Go back to your first drawing. In a second colour, draw the route a secreted protein takes from the gene to the outside of the cell. Number each step. Mark where the protein is made, where it is modified, and how it moves between stations.

In your notebook, head the page **M1-3 Trace the pathway on your cell** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can tell the story out loud without looking, and your numbers are in the right order.

Mini visual sheet

One frame per chunk, shaped for what goes in it. Tall frames are ladders, wide strips are chains, squares are hubs. Draw straight onto this sheet. If a frame is too small for your handwriting, redraw that one in your notes and put the page number on the line.

1 The generalized cell

HUB

2 Permeability

GRID

3 Inside the nucleus

HUB

4 Cytoplasm versus cytosol

SPLIT

5 The secretory pathway

CHAIN

6 Rough ER against smooth ER

GRID

7 The cytoskeleton

HUB

8 9 plus 0 against 9 plus 2

SPLIT